Optimizing CPU and GPU Performance of a RAG Based Chatbot for Minimum-Wage Workers in Metro Manila

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***Abstract*—This study presents the optimization and performance evaluation of *Making Wage Rules Legible*, a retrieval-augmented generation chatbot designed to help minimum-wage workers in Metro Manila access information from Philippine labor documents. The system used Qwen3-1.7B, a Chroma vector database, and three sources covering the Labor Code, Wage Order NCR-27, and statutory monetary benefits. Six configurations were evaluated by varying chunk size, overlap, and top-k while keeping the documents, questions, model, prompt, and generation settings consistent. Each configuration was tested on Google Colab CPU and T4 GPU runtimes using one warm-up run followed by three measured repetitions for three questions of different difficulty levels. The results showed that GPU acceleration substantially reduced processing time, although faster execution did not always produce better-supported answers. Configuration C, using a chunk size of 1,200 characters, an overlap of 150, and top-k of 3, provided the strongest balance. Its average response time decreased from 467.5130 seconds on CPU to 8.1125 seconds on GPU, corresponding to a 57.63-times speedup. It also produced the best observed answer quality, with two correct answers and one partially correct answer. Remaining errors were mainly associated with OCR limitations in the scanned wage-order table and occasional source misinterpretation, showing that retrieval quality and document preparation remain as important as execution speed.**

Keywords*—Retrieval-augmented generation, RAG optimization, chunk size, chunk overlap, vector retrieval, CPU-GPU performance, Qwen3-1.7B, labor information chatbot, minimum-wage workers, statutory monetary benefits.

I. Introduction

Minimum-wage workers often need clear information about working hours, wage rates, holiday pay, and other statutory benefits, yet these rules are commonly presented in lengthy legal documents and technical tables. Although the information is publicly available, locating the correct provision and interpreting it in a practical situation can still be difficult. This concern inspired *Making Wage Rules Legible: A RAG Chatbot for Minimum-Wage Workers in Metro Manila*. The project explores how retrieval-augmented generation can make labor information easier to search by retrieving relevant passages from the Labor Code of the Philippines, Wage Order NCR-27, and the Workers' Statutory Monetary Benefits Handbook before generating an answer. Grounding the response in retrieved evidence is important because workers may rely on the information when understanding their wages and benefits.

Developing a useful RAG chatbot involves more than selecting a language model. Chunk size, overlap, and the number of retrieved passages can influence whether the system finds the correct legal provision or introduces unrelated information. Runtime performance also matters because a technically accurate system may still feel impractical when each response takes several minutes. This study therefore examined six retrieval configurations using the same documents, questions, Qwen3-1.7B model, prompt, and generation settings. Each configuration was evaluated on Google Colab CPU and T4 GPU runtimes, with attention given to indexing time, retrieval time, generation time, total response time, and answer quality. Through this comparison, the study aimed to identify a configuration that provides a thoughtful balance between speed, retrieval relevance, and source-supported answers rather than selecting a setting based on execution time alone.

II. Model Background

The original model proposed for the project was gpt-oss-20b. However, the GPU resources available through Google Colab's free tier were not sufficient for the intended experiment. Since the activity required repeated testing across several retrieval configurations on both CPU and GPU, using a model that could run reliably within the available environment became an important practical consideration. Qwen3-1.7B was therefore selected as a smaller open-source alternative. Its size allowed the complete RAG pipeline to run on a Colab CPU and T4 GPU while still supporting document-grounded question answering.

The language model works together with the retrieval components rather than answering from its internal knowledge alone. Source documents are divided into overlapping chunks and converted into numerical embeddings, which are stored in a Chroma vector database. When a worker submits a question, the same embedding model represents the query and retrieves the most similar

passages. These passages are then included in the prompt given to Qwen3-1.7B. This arrangement helps direct the model toward the supplied labor documents, although it does not guarantee factual correctness. The experiment therefore considered both generated answers and retrieved evidence, particularly because OCR errors, unsuitable chunk boundaries, or irrelevant passages can still influence the model's interpretation. The CPU and GPU tests used the same model and generation settings so that observed performance differences could be associated mainly with the execution device and retrieval configuration.

III. Experimental Methodology

A. Dataset Description

The experiment used three documents related to wage rules and statutory benefits in the Philippines: the Labor Code of the Philippines, Wage Order NCR-27, and the Workers' Statutory Monetary Benefits Handbook 2024 Edition. Together, these sources covered normal working hours, minimum-wage adjustments in Metro Manila, holiday pay, and other employee benefits. The Wage Order was image-based, so optical character recognition was applied before indexing to make its contents searchable. The same OCR-processed file and document-loading procedure were retained throughout all experiments. This consistency mattered because changing the document preparation between runs could influence retrieval results and make the configurations difficult to compare fairly.

B. Chunking Configurations

Six retrieval configurations were evaluated by changing chunk size, chunk overlap, and top-k. Configuration A used 400 characters, an overlap of 50, and top-k 3. Configuration B used 800 characters, an overlap of 100, and top-k 3, while Configuration C used 1,200 characters, an overlap of 150, and top-k 3. Configuration D retained a chunk size of 800 but increased the overlap to 200 to observe the effect of greater contextual repetition. Configuration E repeated the 800-character and 100-overlap setting but increased top-k from 3 to 5. A customized Configuration F was also included using a chunk size of 1,000, an overlap of 200, and top-k 3. All configurations used the same embedding model, Chroma vector database, Qwen3-1.7B model, source documents, and generation settings.

C. Evaluation Method

Three fixed questions representing easy, average, and difficult information needs were used for every configuration. These questions examined the normal eight-hour workday, the two wage rates under Wage Order NCR-27, and the calculation of regular-holiday pay using the first-tranche minimum wage. Each configuration was tested independently on Google Colab CPU and T4 GPU runtimes. One warm-up question was performed before measurement, after which every test question was executed three times. The experiment recorded indexing, retrieval, generation, and total response times, together with generated-token counts and retrieved passages. Answers were reviewed as correct, partially correct, or incorrect by comparing them with predetermined answers and supporting provisions. GPU speedup was calculated by dividing the CPU average time by the corresponding GPU average time. This evaluation considered both efficiency and answer quality, since a faster response was not treated as an improvement when its information was incorrect, incomplete, or unsupported.

IV. Results And Discussion

The experiments showed a clear performance difference between the CPU and T4 GPU runtimes. Across all six configurations, the GPU reduced average response time from several minutes to approximately eight to ten seconds. The amount of improvement varied because each configuration generated answers of different lengths and retrieved different amounts of context. Even with this variation, every configuration produced a GPU speedup greater than one, confirming that the T4 was consistently faster for the tested workload.

TABLE I. Average Response Time And GPU Speedup

Run	*CPU TIme (s)*	*GPU Time(s)*	*GPU Speedup*
A	314.5434	8.7679	35.87x
B	374.0495	8.1394	45.96x
C	467.513	8.1125	57.63x
D	384.8468	8.9699	42.90x
E	615.21	9.7487	63.11x
F	455.9458	8.3925	54.33x

Configuration E achieved the highest calculated response-time speedup at 63.11 times because its CPU execution was particularly slow. However, its GPU response time of 9.7487 seconds was also the slowest among the GPU runs. This observation shows why speedup should be interpreted together with the original execution times. Configuration C recorded the fastest GPU average response at 8.1125 seconds, followed closely by Configuration B at 8.1394 seconds. The small difference between them suggests that response speed alone was not enough to distinguish the more suitable configuration.

TABLE II. INDEXING TIME AND GPU SPEEDUP

Run	*CPU TIme (s)*	*GPU Time(s)*	*GPU Speedup*
A	152.2807	17.2371	8.83x
B	106.7095	15.4651	6.90x
C	90.5029	15.2925	5.92x
D	172.484	15.03	11.48x
E	116.9452	20.5445	5.69x
F	108.9922	15.8611	6.87x

GPU indexing was faster for every configuration, although its improvement was smaller than the response-generation improvement. Configuration D showed the largest indexing speedup at 11.48 times, while Configuration E showed the smallest at 5.69 times. Some variation was expected because document loading and text splitting still involve CPU-bound operations, while Colab resource allocation may also differ between sessions. The recorded measurements were retained as observed instead of being adjusted, since these variations reflect the practical conditions of using the free Colab environment.

TABLE III. ANSWER QUALITY SUMMARY

Run	*CPU Answer Quality*	*GPU Answer Quality*
A	1 correct, 1 partial, 1 incorrect	1 correct, 1 partial, 1 incorrect
B	1 correct, 1 partial, 1 incorrect	1 correct, 1 partial, 1 incorrect
C	1 correct, 1 partial, 1 incorrect	2 correct, 1 partial
D	0 correct, 1 partial, 2 incorrect	0 correct, 1 partial, 2 incorrect
E	0 correct, 2 partial, 1 incorrect	1 correct, 1 partial, 1 incorrect
F	1 correct with a citation issue, 1 partial, 1 incorrect	2 correct with citation issues, 1 partial
A	1 correct, 1 partial, 1 incorrect	1 correct, 1 partial, 1 incorrect
B	1 correct, 1 partial, 1 incorrect	1 correct, 1 partial, 1 incorrect

Answer quality did not improve consistently as chunk size, overlap, or top-k increased. Configuration D showed that increasing overlap from 100 to 200 at a chunk size of 800 did not improve retrieval and led to weaker answers. Configuration E also showed that retrieving five passages instead of three increased response time without resolving the main factual errors. Configuration F produced correct answers for two GPU questions, but its citations were less precise. Configuration C provided the strongest observed quality, with two correct GPU answers and one partially correct answer while also producing the fastest GPU response time.

The average-difficulty wage-order question remained partially correct or incorrect across the configurations. OCR interpreted portions of the scanned wage table incorrectly, causing the model to confuse the ₱60 wage increase with the resulting ₱755 daily wage. This result was particularly meaningful because it showed that retrieval optimization cannot fully correct inaccurate source text. Careful OCR validation and document preparation remain essential when a RAG system is used for legal or wage-related information.

V. RECOMMENDED CONFIGURATION

Based on the combined timing and answer-quality results, Configuration C using a chunk size of 1,200 characters, an overlap of 150 characters, and top-k of 3 is recommended together with the Google Colab T4 GPU runtime. This configuration produced the fastest measured GPU average response time of 8.1125 seconds and completed indexing in 15.2925 seconds. Its average response was approximately 57.63 times faster than its corresponding CPU result. More importantly, it provided the strongest observed answer quality, correctly answering the normal-workday and regular-holiday-pay questions while giving a partially correct response to the wage-order question.

The recommendation was not based on speed alone. Configuration E produced a larger calculated speedup, but retrieving five passages increased its GPU response time and did not improve the factual quality of its answers. Configuration F also performed well, although its generated citations were less precise. Configuration C appeared to provide enough context for the model without introducing as much unrelated or repetitive information. The T4 GPU is therefore recommended for practical use because it reduced response-generation time considerably while remaining available within the tested Colab environment. Nevertheless, the OCR-affected wage-order question should be treated carefully. Improving and validating the OCR text is still necessary before the chatbot can be considered reliable for actual worker guidance.

VI. LIMITATIONS AND POSSIBLE IMPROVEMENTS

A. Limitations

The experiment was conducted using three fixed questions, which provided a controlled comparison but represented only a small portion of the possible questions workers may ask. The most visible limitation came from the

OCR-processed Wage Order NCR-27. Parts of its wage table were recognized incorrectly, causing the model to confuse the ₱60 wage increase with the resulting ₱755 daily minimum wage. Retrieval similarity also did not guarantee that the model interpreted a passage correctly. Some answers used the wrong pay rule, added an amount twice, or cited a passage that did not fully support the conclusion. A few responses reached the generation limit before presenting a final answer. In addition, CPU and GPU resources provided by the free Colab environment varied between sessions, which may have influenced model-loading, indexing, and response times. The experiment used repeated measurements and warm-up runs to reduce this effect, but the runtime conditions could not be controlled completely.

B. Limitations

Future work should begin by validating the OCR output, particularly tables containing wage rates, dates, and monetary values. A cleaner text version or a table-aware OCR method could reduce the retrieval errors observed in the wage-order question. The evaluation set should also be expanded to include more worker-centered questions covering overtime, rest days, special non-working days, leave benefits, and wage deductions. English, Filipino, and Taglish questions could be included to better reflect how workers may communicate in practice. Retrieval may be improved through hybrid search, reranking, metadata filtering, and stronger source-selection rules. The prompt could also require the model to identify the applicable pay category before performing a calculation and to verify that every cited passage supports the final answer. Finally, future experiments could repeat each configuration across several Colab sessions and record one-time model-loading costs separately from normal question-answering time. These improvements would provide a broader and more reliable understanding of the system's practical performance.

VII. Conclusion

This study examined how chunk size, overlap, top-k, and execution device affected the performance of a RAG chatbot for Metro Manila wage and statutory-benefit information. Six configurations were tested using the same source documents, questions, Qwen3-1.7B model, and generation settings. The results showed that the T4 GPU substantially reduced indexing and response-generation times across all configurations. However, the experiments also demonstrated that faster execution did not automatically produce more reliable answers. Retrieval quality, OCR accuracy, and the model's interpretation of the retrieved passages remained important factors.

Configuration C, with a chunk size of 1,200 characters, an overlap of 150, and top-k of 3, provided the best observed balance. On the T4 GPU, it achieved an average response time of 8.1125 seconds and a 57.63-times speedup over its CPU result while producing the strongest answer quality among the tested settings. The remaining error involving the wage-order question was closely connected to incorrect OCR text, which emphasized the importance of preparing and validating source documents before retrieval. Overall, the project showed that a smaller open-source model can support a practical labor-information RAG system within free Colab resources. Still, further document cleaning, expanded evaluation, and stronger evidence validation are needed before the chatbot can be relied upon for real worker guidance.

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